

Drishti: For Airport Visibility Measurement



Drishti at New Delhi Airport

This highly-precise and cost-effective transmissometer developed by Council of Scientific and Industrial Research-National Aerospace Laboratories (CSIR-NAL) is a significant contribution to the field of electronics and airport instrumentation in India

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Visibility on the runway is extremely important for safe landing and take-off of an aircraft. Hence visual-range accessors are mandatory for all airport runways.

Till now the transmissometers used in Indian airports were imported. The requirement of a highly-accurate, cost-effective, low-maintenance, indigenous, state-of-the-art system on par with more-expensive imported systems prompted development of the *Drishti* transmissometer at CSIR-NAL.

Indigenous, economical and rugged

Drishti is the only indigenously-developed transmissometer of its kind available today. It costs just one-third

the price of an imported system, yet gives more precise readings than it.

The device is rugged, has high mean time between failures and is nearly maintenance-free. In fact, the *Drishti* system installed in New Delhi airport in 2011 has never had a failure or maintenance issue to date!

Virtual instrumentation concept with a combination of modular hardware and software design has made serviceability easier, while weather-proof enclosures (IP65) for the electronics hardware make the system

rugged. The availability of complete knowledge base to the component level allows cost-effective servicing and maintenance.

Modern design

Built on an industry-standard FPGA design platform, *Drishti* makes good use of virtual instrumentation concept. Miniaturised, highly-precise, modular optics with provision for manual and automatic alignment has been used in its design.

The use of electronic modulation technique for modulating lamp intensity and synchronous demodulation technique for improving signal-to-noise ratio make *Drishti* highly efficient. It is provided with landline and Wi-Fi communication from runway to air traffic control (ATC) room. It also features web-enabled remote accessibility for data and health monitoring of the system from anywhere in the country. The system facilitates automated measurement of RVR (50-2000 metres) and MOR (10-10,000 metres).

The system

Drishti, developed by CSIR-NAL for Indian Meteorological Department (IMD), meets the specifications for visibility measurement instruments set forth by ICAO and WMO. It has a field-programmable gate array (FPGA)-embedded hardware and web-enabled data acquisition software built in LabView environment.

It basically comprises of a transmitter (a lamp with spectral characteristics similar to runway lights) that sends collimated beam of light and a receiver

What is a transmissometer?

Visual range depends on several environmental factors like fog, snow, rain and storm. It is commonly measured as meteorological optical range (MOR) or runway visibility range (RVR). A transmissometer, also referred to as transmittance meter, is an instrument used to measure the transmission coefficient of the light passing through the atmosphere and thereby determine the MOR and/or RVR.